

BAO CAO THIET KE CSDL QUAN LY CAM DO

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PHAN 1: THIET KE CO SO DU LIEU (3NF)

He thong quan ly cam do duoc thiet ke theo chuan 3NF voi 5 bang chinh, dam bao khong co phu thuoc bat cau va loai bo du thua du lieu.

1.1 Bang KhachHang

Luu thong tin dinh danh cua khach hang cam do.

Cot	Kieu du lieu	Rang buoc	Mo ta
MaKH	INT IDENTITY	PK	Ma khach hang (tu dong tang)
HoTen	NVARCHAR(100)	NOT NULL	Ho ten day du
SoDienThoai	VARCHAR(15)	NOT NULL	So dien thoai lien lac
CCCD	VARCHAR(12)	UNIQUE NOT NULL	Can cuoc cong dan
DiaChi	NVARCHAR(255)		Dia chi thuong tru

1.2 Bang HopDong

Ghi nhan toan bo thong tin moi hop dong vay, bao gom hai moc deadline quan trong de phan dinh giai doan tinh lai don va lai kep.

Cot	Kieu du lieu	Rang buoc	Mo ta
MaHD	INT IDENTITY	PK	Ma hop dong
MaKH	INT	FK	Lien ket KhachHang
NgayVay	DATETIME	NOT NULL	Ngay bat dau vay
SoTienGoc	DECIMAL(18,2)	NOT NULL	Von vay ban dau
Deadline1	DATE	NOT NULL	Han tinh lai don
Deadline2	DATE	NOT NULL	Han thanh ly tai san
TrangThai	NVARCHAR(50)	NOT NULL	Dang vay / Qua han / Da thanh toan / Da thanh ly

1.3 Bang TaiSan

Theo doi tung tai san the chap kem theo trang thai hien tai (dang cam co, da tra khach, san sang thanh ly, da ban).

Cot	Kieu du lieu	Rang buoc	Mo ta
MaTS	INT IDENTITY	PK	Ma tai san
MaHD	INT	FK	Thuoc hop dong nao

TenTaiSan	NVARCHAR(100)	NOT NULL	Ten tai san (xe may, laptop...)
GiaTriDinhGia	DECIMAL(18,2)	NOT NULL	Gia tri uoc luong luc cam
TrangThaiTS	NVARCHAR(50)	NOT NULL	Dang cam co / Da tra / San sang TL / Da ban
MoTa	NVARCHAR(255)		Chi tiet (mau, serial, tinh trang)

1.4 Bang NhanVien

Luu thong tin nhan vien thu tien. Tach rieng de dam bao 3NF (thay vi luu ten nhan vien dang text tho trong Log).

Cot	Kieu du lieu	Rang buoc	Mo ta
MaNV	INT IDENTITY	PK	Ma nhan vien
HoTen	NVARCHAR(100)	NOT NULL	Ho ten nhan vien
SoDienThoai	VARCHAR(15)		So dien thoai
ChucVu	NVARCHAR(50)		Chuc vu (thu quy, giam sat...)

1.5 Bang LogBienDong (Audit Log)

Ghi nhan tung lan khách tra tien. Bang nay la 'nguồn su that duy nhat' cho lich su dong tien, giup truy vet ma khong ghi de tong no.

Cot	Kieu du lieu	Rang buoc	Mo ta
MaLog	INT IDENTITY	PK	Ma log giao dich
MaHD	INT	FK	Thuoc hop dong nao
MaNV	INT	FK	Nhan vien thu tien
NgayGiaoDich	DATETIME	NOT NULL	Thoi diem tra tien
SoTienTra	DECIMAL(18,2)	NOT NULL	So tien khách tra lan nay
DuNoSauKhiTra	DECIMAL(18,2)	NOT NULL	Du no con lai sau giao dich
NoiDung	NVARCHAR(255)		Mo ta (tra lai, tra bot goc, gia han...)

1.6 Quan he giua cac bang

Bang cha	Quan he	Bang con	Y nghia
KhachHang	1 --- N	HopDong	1 KH co nhieu HopDong
HopDong	1 --- N	TaiSan	1 HD co nhieu tai san cam co
HopDong	1 --- N	LogBienDong	1 HD phat sinh nhieu lan tra tien
NhanVien	1 --- N	LogBienDong	1 NV thu tien nhieu lan

1.7 Giai thich chuan hoa 3NF

1NF: Tat ca cac thuoc tinh deu la gia tri don tu. Danh sach tai san duoc tach rieng khoi bang HopDong (khong luu dang danh sach trong 1 o).

2NF: Moi cot khong khoa phu thuoc hoan toan vao khoa chinh cua bang do. Vi du: TenTaiSan phu thuoc vao MaTS (khong phu thuoc mot phan vao MaHD).

3NF: Khong co phu thuoc bat cau. Thong tin KH (HoTen, SDT) nam o bang KhachHang rieng biet, khong lap trong HopDong. Nguoithu duoc tach thanh bang NhanVien rieng thay vi luu text tho trong

PHAN 2: SO DO ERD

So do ERD the hien 5 thuc the chinh va quan he ket noi giua chung. Duoi day la mo ta dang van ban (ERD hình ảnh được xem trong file README.md trên GitHub):

```
[KhachHang] PK: MaKH
HoTen | SoDienThoai | CCCD | DiaChi
|
| 1:N
v
[HopDong] PK: MaHD FK: MaKH
NgayVay | SoTienGoc | Deadline1 | Deadline2 | TrangThai
| |
| 1:N | 1:N
v v
[TaiSan] [LogBienDong] FK: MaHD, MaNV
PK: MaTS FK: MaHD NgayGiaoDich | SoTienTra
TenTaiSan DuNoSauKhiTra | NoiDung
GiaTriDinhGia ^
TrangThaiTS | N:1
MoTa [NhanVien] PK: MaNV
HoTen | SoDienThoai | ChucVu
```

PHAN 3: GIAI THICH THUAT TOAN TINH LAI

He thong su dung co che lai suat linh hoạt 2 giai doan, phan biet theo moc Deadline1. Lai suat áp dụng: 5.000 VND / 1.000.000 VND gọc / ngay (tương đương 0,5%/ngay).

3.1 Giai doan 1: Lai don (tu NgayVay den Deadline1)

CONG THUC LAI DON
$\text{Lai_don} = \text{Goc} \times 0,005 \times \text{So_ngay_giai_doan_1}$
Trong do: $\text{So_ngay_giai_doan_1} = \text{MIN}(\text{TargetDate}, \text{Deadline1}) - \text{NgayVay}$

Ví dụ: Vay 10.000.000 VND trong 20 ngày (chưa qua Deadline1):

$$\begin{aligned}\text{Lai_don} &= 10.000.000 \times 0,005 \times 20 = 1.000.000 \text{ VND} \\ \text{Tong phai tra} &= 10.000.000 + 1.000.000 = 11.000.000 \text{ VND}\end{aligned}$$

3.2 Giai doan 2: Lai kep (tu sau Deadline1 den TargetDate)

Khi hop dong vuot qua Deadline1, lai duoc tinh theo ky thuat lai kep. Co so tinh lai kep la (Goc + Lai don đã tích lũy đến Deadline1), không phải gọc gọc ban đầu.

CONG THUC LAI KEP
$\text{Goc_moi} = \text{Goc_ban_dau} + \text{Lai_don_tai_Deadline1}$
$\text{Tong_no} = \text{Goc_moi} \times (1 + 0,005) ^ n$

$$n = \text{TargetDate} - \text{Deadline1} \quad (\text{so ngay vuot qua Deadline1})$$

Vi du: Vay 10.000.000 VND, Deadline1 = 30 ngay, TargetDate = 45 ngay (n=15 ngay qua han):

$$\text{Lai_don (30 ngay)} = 10.000.000 \times 0,005 \times 30 = 1.500.000 \text{ VND}$$

$$\text{Goc_moi} = 10.000.000 + 1.500.000 = 11.500.000 \text{ VND}$$

$$\text{Tong_no} = 11.500.000 \times (1,005)^{15} = \sim 12.397.000 \text{ VND}$$

3.3 So do thuat toan tong quat (Pseudocode)

```
FUNCTION fn_CalcMoneyContract(ContractID, TargetDate):
    LAY Goc, NgayVay, Deadline1 TU HopDong WHERE MaHD = ContractID

    -- Tinh so ngay tung giai doan
    n1 = DATEDIFF(day, NgayVay, MIN(TargetDate, Deadline1))
    n2 = MAX(0, DATEDIFF(day, Deadline1, TargetDate))

    -- Giai doan 1: Lai don
    Lai_don = Goc * 0.005 * n1

    -- Giai doan 2: Lai kep (chi ap dung neu TargetDate > Deadline1)
    IF n2 > 0 THEN
        Goc_moi = Goc + Lai_don
        Tong_no = Goc_moi * POWER(1.005, n2)
    ELSE
        Tong_no = Goc + Lai_don
    END IF

    -- Tru so tien da tra (tu LogBienDong)
    Da_tra = SUM(SoTienTra) FROM LogBienDong WHERE MaHD = ContractID

    RETURN Tong_no - Da_tra -- Du no con lai
END FUNCTION
```

3.4 Luu y khi cai dat SQL

- Su dung ham POWER(1.005, n) de tinh lai kep, KHONG dung vong lap (hieu suat tot hon).
- DATEDIFF(day, ...) tra ve so nguyen, dam bao khong co sai so floating point.
- Tru so tien da tra (LogBienDong) ra khoi tong no truoc khi tra ve ket qua.
- Xu ly truong hop hop dong da thanh ly (IsSold): khong tinh them lai sau ngay thanh ly.
- fn_CalcMoneyTransaction tinh cho 1 giao dich don le; fn_CalcMoneyContract goi ham nay va tong hop.